Switching Algebra and Its Applications

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Switching Algebra

Basic postulate: existence of two-valued switching variable that takes two distinct values 0 and 1

Switching algebra: algebraic system of set {0,1}, binary operations OR and AND, and unary operation NOT

OR operation	AND operation
$0 + 0 = 0$	$0 . 0 = 0$
$0 + 1 = 1$	$0 . 1 = 0$
$1 + 0 = 1$	$1 . 0 = 0$
$1 + 1 = 1$	$1 . 1 = 1$

NOT operation (complementation): $0' = 1$ and $1' = 0$

OR: also called logical sum
AND: also called logical product

Basic Properties

Idempotency: $x + x = x$

$x \,.\, x = x$

Perfect induction: proving a theorem by verifying every combination of values that the variables may assume

Proof of $x + x = x$: $1 + 1 = 1$ and $0 + 0 = 0$

If x is a switching variable, then: $x + 1 = 1$

$x \,.\, 0 = 0$

$x + 0 = x$

$x \,.\, 1 = x$

Commutativity: $x + y = y + x$

$x \,.\, y = y \,.\, x$

3

Basic Properties (Contd.)

Associativity: $(x + y) + z = x + (y + z)$

$(x \,.\, y) \,.\, z = x \,.\, (y \,.\, z)$

Complementation: $x + x' = 1$

$x \,.\, x' = 0$

Distributivity: $x \,.\, (y + z) = x \,.\, y + x \,.\, z$

$x + y \,.\, z = (x + y) \,.\, (x + z)$

Proof by perfect induction using a truth table:

x	y	z	xy	xz	$y+z$	$x(y+z)$	$xy+xz$
0	0	0	0	0	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	0	1	0	0
0	1	1	0	0	1	0	0
1	0	0	0	0	0	0	0
1	0	1	0	1	1	1	1
1	1	0	1	0	1	1	1
1	1	1	1	1	1	1	1

4

Basic Properties (Contd.)

Principle of Duality:

- Preceding properties grouped in pairs
- One statement can be obtained from the other by interchanging operations OR and AND and constants 0 and 1
- The two statements are said to be dual of each other
- This principle stems from the symmetry of the postulates and definitions of switching algebra w.r.t. the two operations and constants
- Implication: necessary to prove only one of each pair of statements

5

Switching Expressions and Their Manipulation

Switching expression: combination of finite number of switching variables and constants via switching operations (AND, OR, NOT)

- Any constant or switching variable is a switching expression
- If T_1 and T_2 are switching expressions, so are T_1', T_2', T_1+T_2 and T_1T_2
- No other combination of constants and variables is a switching expression

Absorption law: $x + xy = x$

$x(x + y) = x$

Proof: $x + xy = x1 + xy$ [basic property]

$= x(1 + y)$ [distributivity]

$= x1$ [commutativity and basic property]

$= x$ [basic property]

6

Laws of Switching Algebra

Another important law: $x + x'y = x + y$

$x(x' + y) = xy$

Proof: $x + x'y = (x + x')(x + y)$ [distributivity]

$= 1(x + y)$ [complementation]

$= x + y$ [commutativity and basic property]

Consensus theorem: $xy + x'z + yz = xy + x'z$

$(x + y)(x' + z)(y + z) = (x + y)(x' + z)$

Proof: $xy + x'z + yz = xy + x'z + yz1$

$= xy + x'z + yz(x+x')$

$= xy(1 + z) + x'z(1 + y)$

$= xy + x'z$

7

Switching Expression Simplification

Literal: variable or its complement

Redundant literal: if value of switching expression is independent of literal x_i then x_i is said to be redundant

Example: Simplify $T(x,y,z) = x'y'z + yz + xz$

$$
\begin{aligned}
x'y'z + yz + xz &= z(x'y' + y + x) \\
&= z(x' + y + x) \\
&= z(y + 1) \\
&= z1 \\
&= z
\end{aligned}
$$

Thus, literals x and y are redundant in $T(x,y,z)$

Important note: Since no inverse operations are defined in Switching Algebra, cancellations are not allowed

$A + B = A + C$ does not imply $B = C$

Counterexample: $A = B = 1$ and $C = 0$

Similarly, $AB = AC$ does not imply $B = C$

8

De Morgan's Theorems

Involution: $(x')' = x$

De Morgan's theorem for two variables:

$(x + y)' = x' . y'$

$(x . y)' = x' + y'$

Proof by perfect induction:

x	y	x'	y'	$x+y$	$(x+y)'$	$x'y'$
0	0	1	1	0	1	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	0	0

De Morgan's theorems for n variables:

$[f(x_1, x_2, \ldots, x_n, 0, 1, +, .)]' = f(x_1', x_2', \ldots, x_n', 1, 0, ., +)$

9

Simplification Examples

Example: Simplify $T(x,y,z) = (x + y)[x'(y' + z')]' + x'y' + x'z'$

$(x + y)[x'(y' + z')]' + x'y' + x'z'$

$= (x + y)(x + yz) + x'y' + x'z'$

$= (x + xyz + yx + yz) + x'y' + x'z'$

$= x + yz + x'y' + x'z'$

$= x + yz + y' + z'$

$= x + z + y' + z'$

$= x + y' + 1 \quad = 1$

Thus, $T(x,y,z) = 1$, independently of the values of the variables

Example: Prove $xy + x'y' + yz = xy + x'y' + x'z$

$xy + x'y' + yz = xy + x'y' + yz + x'z = xy + x'y' + x'z$

Consensus theorem:

$xy + x'z + yz = xy + x'z$

10

Simplification of Expressions

Example: Simplify $T(x,y,z) = A'C' + ABD + BC'D + AB'D' + ABCD'$

Apply consensus theorem to first three terms -> $BC'D$ is redundant

Apply distributive law to last two terms -> $AD'(B' + BC)$ -> $AD'(B' + C)$
Thus, $T = A'C' + A[BD + D'(B' + C)]$

Consensus theorem:
$xy + x'z + yz = xy + x'z$

Example: Simplify $T(A,B,C,D) = A'B + ABD + AB'CD' + BC$

$A'B + ABD = B(A' + AD) = B(A' + D)$

$AB'CD' + BC = C(B + AB'D') = C(B + AD')$
Thus, $T = A'B + BD + ACD' + BC$

Expand BC to $(A + A')BC$ to obtain $T = A'B + BD + ACD' + ABC + A'BC$

From absorption law: $A'B + A'BC = A'B$

From consensus theorem: $BD + ACD' + ABC = BD + ACD'$
Thus, $T = A'B + BD + ACD'$

11

Switching Functions

Let $T(x_1, x_2, \ldots, x_n)$ be a switching expression:
- Since each variable can assume 0 or 1, 2^n combinations are possible

Determining the value of an expression for an input combination:
Example: $T(x,y,z) = x'z + xz' + x'y'$
$T(0,0,1) = 0'1 + 01' + 0'0' = 1$

Truth table for T

x	y	z	T
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Find the truth table of $T(x,y,z) = x'z + xz' + y'z'$

12

Switching Function (Contd.)

Switching function $f(x_1, x_2, \ldots, x_n)$: values assumed by an expression for all combinations of variables $x_1, x_2, \ldots, x_n$

Complement function: $f'(x_1, x_2, \ldots, x_n)$ assumes value 0 (1) whenever $f(x_1, x_2, \ldots, x_n)$ assumes value 1 (0)

Logical sum of two functions: $f(x_1, x_2, \ldots, x_n) + g(x_1, x_2, \ldots, x_n) = 1$ for every combination in which either f or g or both equal 1

Logical product of two functions: $f(x_1, x_2, \ldots, x_n) \,.\, g(x_1, x_2, \ldots, x_n) = 1$ for every combination for which both f and g equal 1

x	y	z	f	g	f'	$f+g$	fg
0	0	0	1	0	0	1	0
0	0	1	0	1	1	1	0
0	1	0	1	0	0	1	0
0	1	1	1	1	0	1	1
1	0	0	0	1	1	1	0
1	0	1	0	0	1	0	0
1	1	0	1	1	0	1	1
1	1	1	1	0	0	1	0

Canonical Forms

Deriving an expression from a truth table:

- Find the sum of all terms that correspond to combinations for which function is 1
- Each term is a product of the variables on which the function depends
- Variable x_i appears in uncomplemented (complemented) form in the product if has value 1 (0) in the combination

Truth table for $f = x'y'z' + x'yz' + x'yz + xyz' + xyz$

Decimal code	x	y	z	f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

Canonical Sum-of-products (SOPs)

Minterm: a product term that contains each of the *n* variables as factors in either complemented or uncomplemented form

- It assumes value 1 for exactly one combination of variables

Canonical sum-of-products: sum of all minterms derived from combinations for which function is 1

- Also called disjunctive normal expression

Compact representation of switching functions: $\sum(0,2,3,6,7)$

Decimal code	x	y	z	f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

15

Canonical Product-of-sums (POSs)

Maxterm: a sum term that contains each of the *n* variables in either complemented or uncomplemented form

- It assumes value 0 for exactly one combination of variables
- Variable x_i appears in uncomplemented (complemented) form in the sum if it has value 0 (1) in the combination

Canonical product-of-sums: product of all maxterms derived from combinations for which function is 0

- Also called conjunctive normal expression

Compact representation of switching functions: $\prod(1,4,5)$

$$f = (x + y + z')(x' + y + z)(x' + y + z')$$

Decimal code	x	y	z	f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

16

Shannon's Expansion to Obtain Canonical Forms

Shannon's expansion theorem:

$$f(x_1, x_2, \ldots, x_n) = x_1 . f(1, x_2, \ldots, x_n) + x_1' . f(0, x_2, \ldots, x_n)$$
$$f(x_1, x_2, \ldots, x_n) = [x_1 + f(0, x_2, \ldots, x_n)] . [x_1' + f(1, x_2, \ldots, x_n)]$$

Proof by perfect induction: Plug in $x_1 = 1$ and then $x_1 = 0$ to reduce RHS to LHS

Shannon's expansion around two variables:

$$f(x_1, x_2, \ldots, x_n) = x_1x_2f(1, 1, x_3, \ldots, x_n) + x_1x_2'f(1, 0, x_3, \ldots, x_n) + x_1'x_2f(0, 1, x_3, \ldots, x_n) + x_1'x_2'f(0, 0, x_3, \ldots, x_n)$$

Similar Shannon's expansion around all n variables yields the canonical sum-of-products

Repeated expansion of the dual form yields the canonical product-of-sums

17

Simpler Procedure for Canonical SOPs

1. Examine each term: if it is a minterm, retain it; continue to the next term
2. In each product that is not a minterm: check the variables that do not occur; for each x_i that does not occur, multiply the product by $(x_i + x_i')$
3. Multiply out all products and eliminate redundant terms

Example: $T(x,y,z) = x'y + z' + xyz$

$$= x'y(z + z') + (x + x')(y + y')z' + xyz$$
$$= x'yz + x'yz' + xyz' + xy'z' + x'yz' + x'y'z' + xyz$$
$$= x'yz + x'yz' + xyz' + xy'z' + x'y'z' + xyz$$

Canonical product-of-sums obtained in a dual manner

Example:

$$T = x'(y' + z)$$
$$= (x' + yy' + zz')(y' + z + xx')$$
$$= (x' + y + z)(x' + y + z')(x' + y' + z)(x' + y' + z')(x + y' + z)(x' + y' + z)$$
$$= (x' + y + z)(x' + y + z')(x' + y' + z)(x' + y' + z')(x + y' + z)$$

18

Transforming One Form to Another

Example: Find the canonical product-of-sums for

$$T(x,y,z) = x'y'z' + x'y'z + x'yz + xyz + xy'z + xy'z'$$

$$T = (T')' = [(x'y'z' + x'y'z + x'yz + xyz + xy'z + xy'z')']'$$

Complement T' consists of minterms not contained in T. Thus,

$$\begin{aligned} T &= [x'yz' + xyz']' \\ &= (x + y' + z)(x' + y' + z) \end{aligned}$$

Canonical forms are unique

Two switching functions are equivalent if and only if their corresponding canonical forms are identical

19

Functional Properties

Let binary constant a_i be the value of function $f(x_1, x_2, \ldots, x_n)$ for the combination of variables whose decimal code is i. Thus,

$$f(x_1, x_2, \ldots, x_n) = a_0 x_1' x_2' \ldots x_n' + a_1 x_1' x_2' \ldots x_n + \ldots + a_r x_1 x_2 \ldots x_n$$

Factor a_i is set to 1 (0) if the corresponding minterm is (is not) in the canonical form

Since there are 2^n coefficients, each of which can have two values, 0 and 1, there are 2^{2^n} possible switching functions of n variables

Example: Canonical sum-of-products form for two variables

$$f(x,y) = a_0 x'y' + a_1 x'y + a_2 xy' + a_3 xy$$

There are 2^{2^2} functions corresponding to the 16 possible assignments of 0's and 1's to a_0, a_1, a_2, and a_3

20

List of Functions of Two Variables

a_3	a_2	a_1	a_0	$f(x,y)$	*Name of function*	*Symbol*
0	0	0	0	0	Inconsistency	
0	0	0	1	$x'y'$	NOR(dagger)	$x \downarrow y$
0	0	1	0	$x'y$		
0	0	1	1	x'	NOT	x'
0	1	0	0	xy'		
0	1	0	1	y'		
0	1	1	0	$x'y + xy'$	EXCLUSIVE OR (modulo-2 addition)	$x \oplus y$
0	1	1	1	$x' + y'$	NAND(Sheffer stroke)	$x \mid y$
1	0	0	0	xy	AND	$x \cdot y$
1	0	0	1	$xy + x'y'$	Equivalence	$x \equiv y$
1	0	1	0	y		
1	0	1	1	$x' + y$	Implication	$x \rightarrow y$
1	1	0	0	x		
1	1	0	1	$x + y'$	Implication	$y \rightarrow x$
1	1	1	0	$x + y$	OR	$x + y$
1	1	1	1	1	Tautology	

21

The Exclusive-OR Operation

Exclusive-OR: modulo-2 addition, i.e., $A \oplus B = 1$ if either A or B is 1, but not both

Commutativity: $A \oplus B = B \oplus A$

Associativity: $(A \oplus B) \oplus C = A \oplus (B \oplus C) = A \oplus B \oplus C$

Distributivity: $(AB) \oplus (AC) = A(B \oplus C)$

If $A \oplus B = C$, then

$A \oplus C = B$

$B \oplus C = A$

$A \oplus B \oplus C = 0$

Exclusive-OR of an even (odd) number of elements, whose value is 1, is 0 (1)

22

Functionally Complete Operations

Every switching function can be expressed in a canonical form consisting of a finite number of switching variables, constants, and operations +, ., '

A set of operations is functionally complete (or universal) if and only if operations from this set can express every switching function

Example: Set {+, ., '}

Example: Set {+, '} since using De Morgan's theorem, $x \,.\, y = (x' + y')'$.
Thus, + and ' can replace the . in any switching function

Example: Set {., '} for similar reasons

Example: NAND since NAND(x,x) = x' and NAND[NAND(x,y),NAND(x,y)] =xy

Example: NOR since NOR(x,x) = x' and NOR[NOR(x,y),NOR(x,y)] = $x + y$